



East West University
Department of Computer Science and Engineering
Course Outline CSE 200
Spring 2026

Course: CSE200, Computer Aided Engineering Drawing, Sec 5

Credit and Teaching Scheme:

	Theory	Laboratory	Total
Credits	0	1	1
Contact Hours	0	2 Hours/Week for 13 weeks + Final exam in 14 th week	2 Hours/Week for 13 weeks + Final exam in the 14 th week

Prerequisite: None

Instructor Information

Instructor: Tahsin Tariq Banna
 Lecturer, Department of Computer Science and Engineering

E-mail: tahsin.banna@ewubd.edu

Class Routine and Office Hour

Time	Sunday	Monday	Tuesday	Wednesday	Thursday
8:30 - 10:00					
10:00 - 11:50					
11:50 - 1:20			ICE370 AB3-901		ICE370 AB1-402
1:30 - 3:00	Office Hours	CSE200 215 Lab	Office Hours		Office Hours
3:10 - 4:40	Office Hours	Office Hours	CSE438 (3) AB3-701	Office Hours	CSE438 (3) AB3-701
4:50 - 6:20	CSE438 (3) 431 Lab	CSE438 (4) AB2-601	CSE438 (4) 431 Lab	CSE438 (4) AB2-601	

Course Objective

This course introduces students to the standards and conventions of engineering drawing. This course emphasizes on the use of CAD software to generate computer models and technical

drawings. The fundamental principles of orthogonal projections as well as isometric projections and views are also covered in this course.

Knowledge Profile

K3: Theory-based engineering fundamentals; A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline

Learning Domains

Cognitive – **C2:** Understanding **C3:** Applying, **C4:** Analyzing

Psychomotor – **P2:** Manipulation, **P3:** Precision

Affective - **A2:** Responding

Program Outcomes (POs)

PO1: Engineering Knowledge

Complex Engineering Problem Solution

EP1: Depth of knowledge required

EP2: Range of conflicting requirements

Course Outcomes (Cos)

After completion of this course students will be able to:

CO	CO Description	PO	Learning Domains	Knowledge Profile	Complex Engineering Problem Solving
CO1	Understand the fundamental concepts of engineering drawings and use computer aided drawing instruments to create geometric constructions.	PO1	C2, C3	K3	
CO2	Understand the theory of projection and sectional views. Use CAD tools to apply and examine multi-view projection techniques i.e. orthographic projections of lines, planes and solids.	PO1	C2, C3, C4	K3	
CO3	Apply and examine isometric drawing techniques, isometric projections and conversion among 3D drawing, isometric drawing and multi-view projections using CAD tools.	PO1	C3, C4	K3	
CO4	Demonstrate skills on Computer Aided Design instruments to create 2D and 3D engineering drawings, write report on a complete design.	PO1	C3, P2, P3, A2	K3	EP1, EP2

Course Contents, Teaching-Learning Method and Assessment Scheme

Course Topic	Teaching - Learning Method	CO	Mark of Cognitive Learning Levels			Mark of COs	Exam Mark
			C2	C3	C4		
Introduction to drawing instruments, Lines, Lettering and Dimensioning.	Lecture, Demonstration, Class Discussion, Discussion with Instructor/TA	CO1	5			5	Mid Semester Assessment (20)
Scales: Plain, Diagonal and Vernier Scale. Curves used in engineering practice: ellipse, parabola, hyperbola.	Do	CO1	5			5	
Projection of Lines: Oblique Lines, Traces. Applications of Projection of Lines. Projection of Planes: Polygonal Lamina, Circular Lamina. Projection of Solids: Cube, Prism, Pyramid, Cylinder, Cone. Suspended Solids	Do	CO1	5			5	
Sectional views: Full section, Half section, Offset section, Broken section	Do	CO2	5			5	
Isometric Projections: Isometric Lines, Planes and Scale. Conversion of given 2D views to Isometric Projection/View. Conversion of given 3D View to 2D representation	Do	CO3			10	10	Final Exam (30)

Drafting (3D): Isometric drawing	Do	CO3			10	10	Lab Performance (30)
Drafting (3D): 3D modelling	Do	CO3			10	10	
Orthographic Projection: First and Third angle projection methods	Do	CO2			10	10	
Drafting (2D): Drawing 2D geometric constructions, Floor planning	Do	CO2			10	10	
Drafting (Projection): Drawing multi-view projections and sectional views.	Do	CO2			10	10	

Mini Project

Mini Project	Teaching-Learning Method	CO	EP/EA	Mark of Cognitive Learning Level		Mark of Psychomotor Learning Levels		Mark of Affective Learning Level	CO Mark
				C2		P2	P3	A2	
Lab-based Mini Project On Drafting (3d modelling) including Report and Presentation	Group based moderately complex 3d modelling project with report writing	CO4	EP1, EP2	7		3	3	2	15

Overall Assessment Scheme

Assessment Area	CO				Other	PO Marks
	CO1	CO2	CO3	CO4		PO1
Attendance						
Assignment				10		10
Midterm Semester Assessment	15	5				20
Lab Final Exam			30			30
Lab Performance		20				20
Mini Project				20		20
Total Mark	15	25	30	30		100

Teaching Materials/Equipment

Textbook:

1. Engineering Drawing (With an introduction to AutoCAD) by Dhananjay A Jolhe
2. Engineering Drawing by Basant Agrawal & C M Agrawal

Lab Manual:

Lab manual will be provided.

Project Description:

Project description will be provided.

Equipment/Software:

AutoCad, Fusion360, SolidWorks, OnShape

Grading System

Marks (%)	Letter Grade	Grade Point
80% and above	A+	4.00
75% to less than 80%	A	3.75
70% to less than 75%	A-	3.50
65% to less than 70%	B+	3.25
60% to less than 65%	B	3.00
55% to less than 60%	B-	2.75
50% to less than 55%	C+	2.50

45% to less than 50%	C	2.25
40% to less than 45%	D	2.00
Less than 40%	F	0.00

Exam Dates

*As per Academic calendar

Academic Code of Conduct

Academic Integrity:

Any form of cheating, plagiarism, personification, falsification of a document as well as any other form of dishonest behavior related to obtaining academic gain or avoidance of evaluative exercises committed by a student is an academic offence under Academic Code of conduct and **may lead to severe penalties as decided by the Disciplinary committee of the university.**

Special Instructions:

- Students are expected to attend all the classes and examinations. A student **MUST** have at least 80% class attendance to sit for final exam.
- Students will not be allowed to enter into the classroom after 20 minutes of the starting time.
- For plagiarism, Grade will automatically become zero for that exam/assignment.
- Normally there will be **NO make-up exam**. However, in case of **severe illness, death of any family member, any family emergency, or any humanitarian ground**, if a student misses any exam, he/she **MUST** get approval of makeup exam by written application to the Chairperson through the Course Instructor **within 48 hours** of the exam time. Proper supporting documents in favor of the reason of the missing the exam have to be presented with the application.
- For **Final exam**, there will be No makeup exam. However, in case of **severe illness, death of any family member, any family emergency, or any humanitarian ground**, if a student misses the final exam, he/she **MUST** get approval of **Incomplete Grade** by written application to the Chairperson through the Course Instructor **within 48 hours** of the final exam time. Proper supporting documents in favor of the reason of the missing the final exam have to be presented with the application. **It is the responsibility of the student to arrange an Incomplete Exam within the deadline mentioned in the Academic Calendar in consultation with the course Instructor.**
- All mobile phones **MUST** be turned into **Silent Mode** during class and exam period.
- There is **zero tolerance for cheating** in exam. Students caught with cheat sheets in their possession, whether used or not; writing on the palm of hand, back of calculators, chairs or nearby walls; copying from cheat sheets or other cheat sources; copying from other examinee etc. would be treated as cheating in the exam hall. The only penalty for cheating is **expulsion for several semesters as decided by the Disciplinary Committee of the university.**